SANJIT MATHUR

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Education & Summary

University of Wollongong in Dubai
BE Computer and Autonomous Systems Engineering (Expected 2027)

Skills

Languages: Python, TypeScript, SQL, C++
AI/ML/Data: XGBoost, LightGBM, Scikit-learn, Pandas, NumPy, Logistic Regression, LSTM, SHAP
Backend: FastAPI, Node.js, Express.js, REST APIs, Pydantic, PostgreSQL
Infrastructure & Tools: Docker, Kubernetes, Streamlit, Plotly

Work Experience

AI Engineering Intern | Publicis Sapient, Dubai (July 2026 – Present)

- Engineered a production-grade REST API using Node.js, Express 5, TypeScript, and PostgreSQL, implementing layered architecture, Prisma ORM, relational data models, and full CRUD task management with pagination and filtering.
- Built a secure authentication and authorization system with JWT, bcrypt, RBAC, and reusable middleware, enforcing role-based permissions, resource ownership validation, and admin overrides across protected endpoints.
- Strengthened API reliability and maintainability with Zod request validation, centralized error handling, Jest + Supertest testing, ESLint, Prettier, and tsx, delivering structured error responses and a robust developer workflo

AI Engineering Intern | Baraka Financial Ltd., Dubai (Feb 2026 – April 2026)

- Architected and shipped 5 production microservices (Position Search, Trading Account Monitor, CRA, EOD History, Slack Automation) deployed to Kubernetes; used daily by entire operations team for portfolio management and trade reconciliation.
- Built LLM-powered error classification pipeline that reduced manual support triage time by 6+ hours per week, automated categorization of support tickets across OMS, Instruments, and Wallet microservices.
- Engineered unified portfolio state model consolidating inconsistent data schemas across distributed microservices; eliminated data discrepancies that previously affected user accounts, enabling accurate cross-market position reporting.

Digital Intern | IndiGo InterGlobe Aviation Ltd., Gurgaon (August 2025 – Sept 2025) – GitHub

- Engineered end-to-end Logistic Regression pipeline for on-time arrival prediction (DEL–BOM sector), achieved 88% accuracy using 1,000 flight records and 30-variable dataset.
- Conducted comprehensive feature engineering: block-hour overrun computation, departure

  delay anomaly detection, ATC congestion indexing, weather factor extraction; identified top 6 features via correlation analysis across 10 numerical variables and 3 categorical aircraft types.

Projects

Distributed Fraud Detection & Anomaly Analysis | GitHub, Live Website

- Built ensemble anomaly detection system (Isolation Forest, LOF, Autoencoder) processing 284K credit card transactions..
- Engineered domain-specific features and trained models on normal-only data; automated threshold optimization via precision-recall analysis.
- Developed interactive Streamlit dashboard with SHAP explainability for transaction-level fraud reasoning.

Multi Domain Demand Forecaster | GitHub, Live Website

- Built ensemble ML forecasting engine using XGBoost + LightGBM stacking with Ridge meta-learner for multi-domain demand prediction (airline bookings, e-commerce, payment volume).
- Engineered advanced temporal features: cyclical encodings, rolling statistics for trend/seasonality capture.
- Deployed FastAPI prediction microservice and Streamlit analytics dashboard for real-time forecasts and model performance monitoring.

ExamForge — AI Exam Generation Platform | GitHub

- Built full-stack LLM-powered exam generation platform enabling teachers to dynamically create multi-format assessments.
- Designed interactive workflows for exam generation, student attempts, and automated evaluation.

Orvyn ExoArm – Rehabilitation Exoskeleton (Current – May 2026)

- Developing signal classification pipeline to differentiate between intended finger movements and noise, targeting **90%+ accuracy** for assisted rehab exercises.
- Designing a microcontroller-based rehabilitation exoskeleton that interprets sEMG signals to assist finger movement through real-time signal processing.